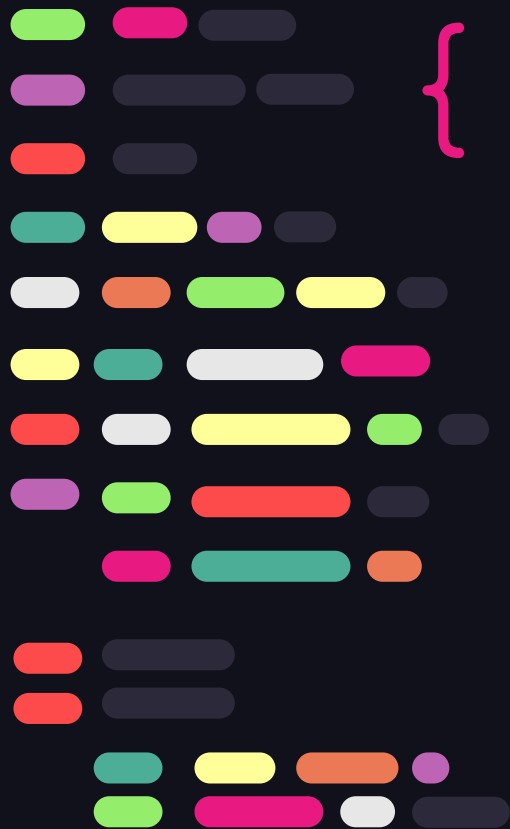




Laboratorio Forense y Adquisición Digital

< Introducción a la Forensia Digital >





01 { ..

Laboratorio

Forense Digital

< IFD >



{ [Laboratorio Forense]



< Estación de trabajo forense >

< Bloqueadores de Escritura >

< Duplicadores / Clonadores >

< Tarjetas / adaptadores >

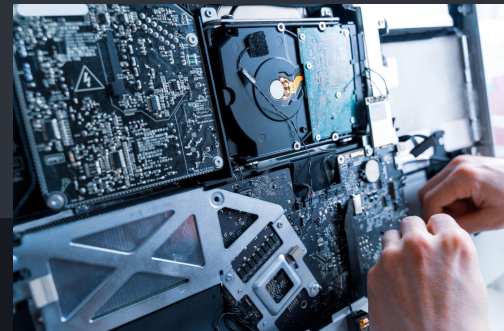
< Dispositivos móviles y accesorios >

< Almacenamiento seguro / respaldo >

< Captura de red >

< Cámaras >

< Infraestructura >



{ [Laboratorio Forense]

Distribuciones Forenses

Herramientas para:

- Adquisición
- Análisis de sistemas de archivos / discos
- Análisis de memoria
- Análisis móvil
- Análisis de tráfico de red
- Reporte / timeline / visualización

} Software de aseguramiento de evidencia,
comprobación de hash y validación



{ [Buenas Prácticas]

- Verificar que los equipos y software cumplen estándares aceptados (por ejemplo NIST, ENFSI, ISO/IEC 17025 si corresponde).
- Mantener cadena de custodia documentada.
- Generar hashes antes y después de adquisiciones, de copias, etc.
- Aislamiento del medio original, trabajar siempre con copias.
- Documentación de todo lo que se hace (qué software versiones, qué hardware, configuración, etc.).
- Control de acceso físico al laboratorio.
- Actualización de herramientas, revisión de integridad de software.
- Entrenamiento del personal.



{ [Links]

Distribuciones Forenses completas

Kali Linux Modo Forense /

<https://www.kali.org/docs/general-use/kali-linux-forensics-mode/>

CAINE Linux / <https://www.caine-live.net/page5/page5.html>

Bloqueadores de Escritura

<https://dataexpert.eu/products/forensic-hardware-logicube/writeprotect>

}





02 { ..

Adquisición

Digital

< IFD >



Adquisición < /1 > {



< Es el proceso de obtener y preservar evidencia digital en un entorno forense >

}

Conservación < /2 > {



< Garantizar la integridad y cadena de custodia >

}



Orden de volatilidad < /3 > {



< secuencia en la que se deben recopilar los datos en función de su volatilidad o probabilidad de cambio >

}

< 1. Registros y caché de la CPU >

< 2. Tablas de enrutamiento, caché ARP, tabla de procesos, estadísticas del kernel, **RAM** >



< 3. Sistemas de archivos temporales >

< 4. Disco (HDD, SSD, NVM, ...)

< 5. Datos de registro y monitoreo remotos >

< 6. Configuración física y topología de red >

< 7. Medios de almacenamiento externo >





Orden de volatilidad < /3 > {

Ordenar los elementos de datos
forenses según su volatilidad,
de mayor a menor:

Archivos Temporales del Sistema
Logs de Servidor Syslog
Registros de la CPU
Contenido de la RAM
Pendrive conexión USB3
Tabla de Procesos
Diagrama de Topología de Red
Tabla de Ruteo
SSD

}

Orden	Elemento de Dato
1	
2	
3	
4	
5	
6	
7	
8	
9	





Orden de volatilidad < /3 > {

Ordenar los elementos de datos forenses según su volatilidad, de mayor a menor:

Archivos Temporales del Sistema
Logs de Servidor Syslog
Registros de la CPU
Contenido de la RAM
Pendrive conexión USB3
Tabla de Procesos
Diagrama de Topología de Red
Tabla de Ruteo
SSD

}

Orden	Elemento de Dato
1	Registros de la CPU
2	Tabla de Ruteo
3	Tabla de Procesos
4	Contenido de la RAM
5	Archivos Temporales del Sistema
6	SSD
7	Logs de Servidor Syslog
8	Diagrama de Topología de Red
9	Pendrive conexión USB3



{ [Métodos para Adquirir Datos de forma Off-line]

Creación de imágenes de disco

< Es el proceso de crear una copia **bit a bit** de un disco, es una réplica **completa** de los datos originales >

< Permite examinar los datos sin alterar la evidencia original >



Bloqueador de Escritura Hardware



Bloqueador de Escritura Hardware

< WriteProtect™ PORTABLE >



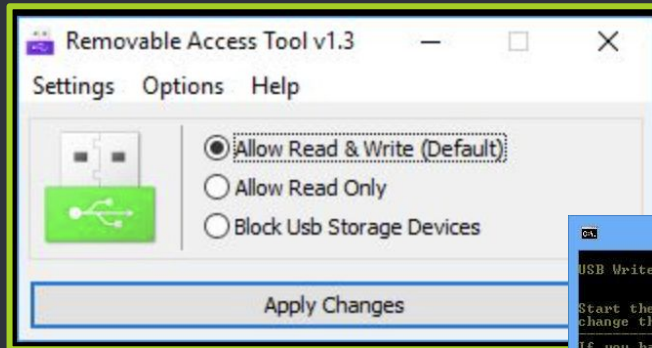
< WriteProtect™ DESKTOP >



< WriteProtect™ BAY >



Bloqueador de Escritura Software



{ [Almacenamiento Seguro de los datos adquiridos]

< Una vez que se haya adquirido los datos
necesarios, se deben almacenar de forma
segura para evitar la manipulación y
mantener su integridad >

}



1 { [Garantizar **integridad** y
2 Cadena de custodia
3 de la evidencia digital]
4
5
6
7
8

Cadena de Custodia

9
10 < Mantener una cadena de custodia
11 garantiza que la evidencia no haya sido
12 alterada → **Integridad** >
13 }
14



```
1 { [Garantizar integridad
2
3
4   de la evidencia digital]
5
6
7   < Documentar cada paso >
8   < Transporte seguro >
9   < Hashing >
10  < Bloqueo de Escritura >
11
12 }
13
14
```





[Cadena de Custodia]


 PROVINCIA DE BUENOS AIRES
 PODER JUDICIAL
 MINISTERIO PÚBLICO

Cadena Nro. :

DOCUMENTO DE CADENA DE CUSTODIA

UFI: _____ DEPTO. JUDICIAL: _____
 I.P.P./CAUSA: _____
 DEP. POLICIAL INTERVINIENTE: _____
 CARATULA: _____
 VÍCTIMA: _____
 IMPUTADO: _____

DATOS DE LA MUESTRA

Descripción de la muestra:

Modo de conservación: (tachar lo que NO corresponda)	0°C a -4°C / freezer / lugar frío o seco / Otro:			
Lugar de toma / Identidad de la muestra:				
Perito que intervino: <u>(PRIMER ESLABÓN</u> de la cadena de custodia)	Datos personales		Firma	
Fecha / Hora:				
OBSERVACIONES:				

Fecha D/M/A	Hora	Datos personales Legajo/DNI, Apellido y Nombre	Dependencia- Entidad	RESPONSABLES EN LA CADENA DE CUSTODIA (FIRMA)
2				

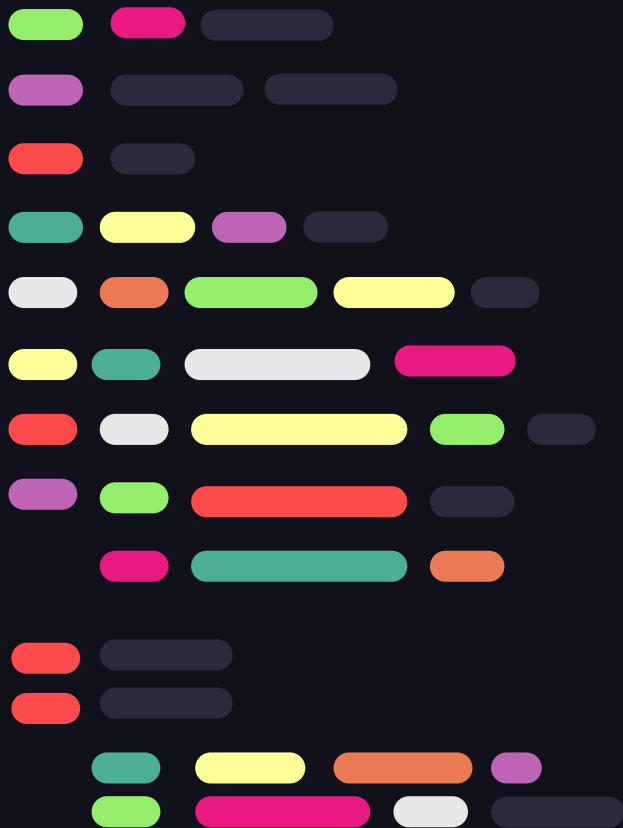


{ [Herramientas Forenses para la creación de imágenes de disco]

- < dd y dc3dd >
- < Guymager >
- < FTK Imager >

}





dc3dd}

< Adquisición forense en Linux >



{ [Previo a la adquisición]

< df >

```
(nany@kali-fusion)-[~]  
$ df -h
```

Filesystem	Size	Used	Avail	Use%	Mounted on
udev	2.1G	0	2.1G	0%	/dev
tmpfs	442M	1.3M	441M	1%	/run
/dev/nvme0n1p2	73G	23G	47G	33%	/
tmpfs	2.2G	4.0K	2.2G	1%	/dev/shm
tmpfs	5.0M	0	5.0M	0%	/run/lock
tmpfs	2.2G	448K	2.2G	1%	/tmp
/dev/nvme0n1p1	512M	160K	512M	1%	/boot/efi
tmpfs	442M	112K	442M	1%	/run/user/1000



{ [Previo a la adquisición]

< lsblk >

```
(nany@kali-fusion)-[~]  
$ lsblk -a
```

NAME	MAJ:MIN	RM	SIZE	RO	TYPE	MOUNTPPOINTS
sda	8:0	1	0B	0	disk	
sdb	8:16	1	3.7G	0	disk	
└─sdb1	8:17	1	3.7G	0	part	
sr0	11:0	1	1024M	0	rom	
nvme0n1	259:0	0	76G	0	disk	
└─nvme0n1p1	259:1	0	512M	0	part	/boot/efi
└─nvme0n1p2	259:2	0	74.5G	0	part	/
└─nvme0n1p3	259:3	0	976M	0	part	[SWAP]



{ [Previo a la adquisición]

< blkid >

```
(nany@kali-fusion)-[~]  
$ sudo blkid /dev/sdb1  
/dev/sdb1: BLOCK_SIZE="512" UUID="7EFF11B23B4527FA" TYPE="ntfs" PARTUUID="aee14070-01"
```

}

```
(nany@kali-fusion)-[~]  
$ sudo blkid /dev/nvme0n1p2  
/dev/nvme0n1p2: UUID="65fd04e1-e1ef-4e03-9660-4e19984fdc2e" BLOCK_SIZE="4096" TYPE="ext4"  
PARTUUID="0401df82-b045-4e3f-ab57-a573e4a57789"
```



{ [dc3dd / Adquisición .img]

< dc3dd >

```
(nany@kali-fusion)-[~/Desktop/dispositivoUSB]
$ sudo dc3dd if=/dev/sdb1 of=dispositivoUSB.img log=log_dispositivoUSB.txt

dc3dd 7.2.646 started at 2025-10-02 12:35:12 -0300
compiled options:
command line dc3dd if=/dev/sdb1 of=dispositivoUSB.img log=log_dispositivoUSB.txt
device size: 7829504 sectors (probed),    4,008,706,048 bytes
sector size: 512 bytes (probed)
█ 204046336 bytes ( 195 M ) copied ( 5% ),    11 s, 18 M/s
```



{ [dc3dd / Adquisición .img]

< dc3dd >

```
(nany@kali-fusion)-[~/Desktop/dispositivoUSB]
$ sudo dc3dd if=/dev/sdb1 of=dispositivoUSB.img log=log_dispositivoUSB.txt

dc3dd 7.2.646 started at 2025-10-02 12:35:12 -0300
compiled options:
command line dc3dd if=/dev/sdb1 of=dispositivoUSB.img log=log_dispositivoUSB.txt
device size: 7829504 sectors (probed),    4,008,706,048 bytes
sector size: 512 bytes (probed)
4008706048 bytes ( 3.7 G ) copied ( 100% ),  223 s, 17 M/s

input results for device `/dev/sdb1':
  7829504 sectors in
  0 bad sectors replaced by zeros

output results for file `dispositivoUSB.img':
  7829504 sectors out

dc3dd completed at 2025-10-02 12:38:55 -0300
```



{ [dc3dd / Adquisición .img]

< dc3dd >

```
(nany@kali-fusion)-[~/Desktop/dispositivoUSB]
$ ls -alh dispositivoUSB.img
-rw-r--r-- 1 root root 3.8G Oct  2 12:38 dispositivoUSB.img
```

```
(nany@kali-fusion)-[~/Desktop/dispositivoUSB]
$ sudo lsblk -l
[sudo] password for nany:
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda          8:0    1   0B  0 disk
sdb          8:16    1  3.7G  0 disk
sdb1         8:17    1  3.7G  0 part
sr0         11:0    1 1024M  0 rom
nvme0n1     259:0    0   76G  0 disk
nvme0n1p1   259:1    0  512M  0 part /boot/efi
nvme0n1p2   259:2    0 74.5G  0 part /
nvme0n1p3   259:3    0  976M  0 part [SWAP]
```



{ [dc3dd / Adquisición .img]

< comprobación de integridad >

```
(nany@kali-fusion)-[~/Desktop/dispositivoUSB]
$ sudo sha256sum dispositivoUSB.img
d6af86f3e54b3e8d05a6faa9f51a44b51e033a1509ab7549059abb60af684791  dispositivoUSB.img

(nany@kali-fusion)-[~/Desktop/dispositivoUSB]
$ sudo sha256sum /dev/sdb1
d6af86f3e54b3e8d05a6faa9f51a44b51e033a1509ab7549059abb60af684791  /dev/sdb1
```



}



{ [dc3dd / Adquisición .dd]

< dc3dd >

```
(nany@kali-fusion)-[~/Desktop/dispositivoUSB]
$ sudo dc3dd if=/dev/sdb of=dispositivoUSB_3.dd hash=sha256 log=dispositivoUSB_3_log.txt

dc3dd 7.2.646 started at 2025-10-02 15:12:38 -0300
compiled options:
command line dc3dd if=/dev/sdb of=dispositivoUSB_3.dd hash=sha256 log=dispositivoUSB_3_log.txt
device size: 7831552 sectors (probed),    4,009,754,624 bytes
sector size: 512 bytes (probed)
█ 1395818496 bytes ( 1.3 G ) copied ( 35% ),    74 s, 18 M/s
```



{ [dc3dd / Adquisición .dd]

< dc3dd >

```
(nany@kali-fusion)-[~/Desktop/dispositivoUSB]
$ sudo dc3dd if=/dev/sdb of=dispositivoUSB_3.dd hash=sha256 log=dispositivoUSB_3_log.txt

dc3dd 7.2.646 started at 2025-10-02 15:12:38 -0300
compiled options:
command line dc3dd if=/dev/sdb of=dispositivoUSB_3.dd hash=sha256 log=dispositivoUSB_3_log.txt
device size: 7831552 sectors (probed),    4,009,754,624 bytes
sector size: 512 bytes (probed)
  4009754624 bytes ( 3.7 G ) copied ( 100% ),  223 s, 17 M/s

input results for device `/dev/sdb':
  7831552 sectors in
  0 bad sectors replaced by zeros
  aecdd444fbc1bed12056833b78ebdeb36ab32390b84bbb8f083d88532ccff720 (sha256)

output results for file `dispositivoUSB_3.dd':
  7831552 sectors out
dc3dd completed at 2025-10-02 15:16:20 -0300
```



{ [dc3dd / Adquisición .dd]

< dc3dd >

```
(nany@kali-fusion)-[~/Desktop/dispositivoUSB]
$ ls -lha dispositivoUSB_3.dd
-rw-r--r-- 1 root root 3.8G Oct  2 15:16 dispositivoUSB_3.dd
```

```
(nany@kali-fusion)-[~/Desktop/dispositivoUSB]
$ sudo lsblk -l
```

NAME	MAJ:MIN	RM	SIZE	RO	TYPE	MOUNTPOINTS
sda	8:0	1	0B	0	disk	
sdb	8:16	1	3.7G	0	disk	
sdb1	8:17	1	3.7G	0	part	
sr0	11:0	1	1024M	0	rom	
nvme0n1	259:0	0	76G	0	disk	
nvme0n1p1	259:1	0	512M	0	part	/boot/efi
nvme0n1p2	259:2	0	74.5G	0	part	/
nvme0n1p3	259:3	0	976M	0	part	[SWAP]

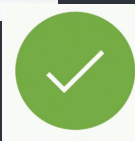


{ [dc3dd / Adquisición .dd]

< comprobación de integridad >

```
(nany@kali-fusion)-[~/Desktop/dispositivoUSB]
$ sudo sha256sum dispositivoUSB_3.dd
aecdd444fbc1bed12056833b78ebdeb36ab32390b84bbb8f083d88532ccff720  dispositivoUSB_3.dd

(nany@kali-fusion)-[~/Desktop/dispositivoUSB]
$ sudo sha256sum /dev/sdb
aecdd444fbc1bed12056833b78ebdeb36ab32390b84bbb8f083d88532ccff720  /dev/sdb
```



}



{ [Montar imagen]

< Ver particiones y offsets >

```
(nany@kali-fusion)-[~/Desktop/dispositivoUSB]
$ fdisk -l dispositivoUSB_3.dd
Disk dispositivoUSB_3.dd: 3.73 GiB, 4009754624 bytes, 7831552 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xae14070
```

Device	Boot	Start	End	Sectors	Size	Id	Type
dispositivoUSB_3.dd1		2048	7831551	7829504	3.7G	7	HPFS/NTFS/exFAT

}

La partición 1 empieza en el sector **2048**



```
1 { [Montar imagen]
```

```
2  
3  
4  
5  
6 < Calcular el offset en bytes >
```

```
7  
8 offset = Sector Inicio * Tamaño Sector  
9 (normalmente 512 bytes)
```

```
10  
11 Offset = 2048 * 512 = 1048576
```

```
12 }  
13  
14
```



```
1 { [Montar imagen]
```

```
2  
3  
4  
5  
6  
7 < Montar usando offset >
```

```
8  
9 sudo mount -o loop,offset=1048576  
10 dispositivoUSB_3.dd /mnt/dispositivoUSB  
11
```

```
12 }  
13  
14
```



{ [Montar imagen]

< Montar usando offset >

```
(nany@kali-fusion)-[~/Desktop/dispositivoUSB]
$ sudo mount -o loop,offset=1048576 dispositivoUSB_3.dd /mnt/dispositivoUSB
[sudo] password for nany:
```

```
(nany@kali-fusion)-[~/Desktop/dispositivoUSB]
$ ls -lha /mnt/dispositivoUSB
total 1.2G
drwxrwxrwx 1 root root 8.0K Sep 30 13:03 .
drwxr-xr-x 3 root root 4.0K Oct 2 13:05 ..
-rwxrwxrwx 1 root root 27K Sep 30 09:40 345.pdf
-rwxrwxrwx 1 root root 54K Sep 30 08:50 375.pdf
-rwxrwxrwx 2 root root 991M Jun 15 19:24 CellebriteReader.exe
-rwxrwxrwx 2 root root 1.5M Sep 30 12:49 IMG_1286.jpeg
-rwxrwxrwx 2 root root 2.3M Sep 30 13:03 IMG_6652.HEIC
-rwxrwxrwx 2 root root 6.7K Sep 30 08:53 'Nuevo Documento de texto.txt'
-rwxrwxrwx 1 root root 1.9M Sep 30 12:11 Ratool.tar
-rwxrwxrwx 2 root root 167M Sep 30 09:10 SysinternalsSuite.zip
drwxrwxrwx 1 root root 0 Jan 2 2023 'System Volume Information'
-rwxrwxrwx 1 root root 244 Sep 30 12:14 activity.log
-rwxrwxrwx 2 root root 0 Sep 30 09:21 bulk_extractor-1.5.0-windowsinstaller.exe
-rwxrwxrwx 1 root root 0 Sep 30 09:19 exploit.bin
-rwxrwxrwx 2 root root 393 Sep 30 12:12 exploit_readme.txt
-rwxrwxrwx 2 root root 90 Sep 30 12:13 exploittttttt.bin
-rwxrwxrwx 2 root root 2.8M Sep 30 12:55 f230030702.jpg
-rwxrwxrwx 2 root root 180K Sep 30 12:11 hashmyfiles-x64.tar
-rwxrwxrwx 1 root root 2.3M Sep 30 12:06 imagen.png
-rwxrwxrwx 2 root root 347 Sep 30 12:15 rule.yara
```



{ [Desmontar imagen]

```
(nany@kali-fusion)-[~/Desktop/dispositivoUSB]
$ sudo umount /mnt/dispositivoUSB

(nany@kali-fusion)-[~/Desktop/dispositivoUSB]
$ ls -lha /mnt/dispositivoUSB
total 8.0K
drwxr-xr-x 2 root root 4.0K Oct  2 13:05 .
drwxr-xr-x 3 root root 4.0K Oct  2 13:05 ..

(nany@kali-fusion)-[~/Desktop/dispositivoUSB]
$ █
```



{ [Montar imagen]

< .img >

```
(nany@kali-fusion)-[~/Desktop/dispositivoUSB]
$ df -h
```

Filesystem	Size	Used	Avail	Use%	Mounted on
udev	2.1G	0	2.1G	0%	/dev
tmpfs	442M	1.3M	441M	1%	/run
/dev/nvme0n1p2	73G	26G	44G	38%	/
tmpfs	2.2G	4.0K	2.2G	1%	/dev/shm
tmpfs	5.0M	0	5.0M	0%	/run/lock
tmpfs	2.2G	448K	2.2G	1%	/tmp
/dev/nvme0n1p1	512M	160K	512M	1%	/boot/efi
tmpfs	442M	112K	442M	1%	/run/user/1000

```
(nany@kali-fusion)-[~/Desktop/dispositivoUSB]
$ sudo mkdir -p /mnt/dispositivoUSB

(nany@kali-fusion)-[~/Desktop/dispositivoUSB]
$ sudo mount -o loop dispositivoUSB.img /mnt/dispositivoUSB
```

```
(nany@kali-fusion)-[~/Desktop/dispositivoUSB]
$ df -h
```

Filesystem	Size	Used	Avail	Use%	Mounted on
udev	2.1G	0	2.1G	0%	/dev
tmpfs	442M	1.3M	441M	1%	/run
/dev/nvme0n1p2	73G	26G	44G	38%	/
tmpfs	2.2G	4.0K	2.2G	1%	/dev/shm
tmpfs	5.0M	0	5.0M	0%	/run/lock
tmpfs	2.2G	464K	2.2G	1%	/tmp
/dev/nvme0n1p1	512M	160K	512M	1%	/boot/efi
tmpfs	442M	112K	442M	1%	/run/user/1000
/dev/loop0	3.8G	1.2G	2.6G	32%	/mnt/dispositivoUSB



{ [Listar Contendio]

< dc3dd >

```
(nany@kali-fusion)-[~/Desktop/dispositivoUSB]
$ ls /mnt/dispositivoUSB
345.pdf          activity.log
375.pdf          bulk_extractor-1.5.0-windowsinstaller.exe
CellebriteReader.exe  exploit.bin
IMG_1286.jpeg    exploit_readme.txt
IMG_6652.HEIC    exploittttttt.bin
'Nuevo Documento de texto.txt' f230030702.jpg
Ratool.tar       hashmyfiles-x64.tar
SysinternalsSuite.zip imagen.png
'System Volume Information' rule.yara
```



{ [Desmontar Imagen]

< dc3dd >

```
(nany@kali-fusion)-[~/Desktop/dispositivoUSB]
$ sudo umount /mnt/dispositivoUSB
[sudo] password for nany:

(nany@kali-fusion)-[~/Desktop/dispositivoUSB]
$ ls /mnt/dispositivoUSB

(nany@kali-fusion)-[~/Desktop/dispositivoUSB]
$ █
```



{ [macOS Sequoia versión 15.6.1]

```
sandrazilla@Mac ~ % diskutil list
/dev/disk0 (internal, physical):
#:

| #: | TYPE                  | NAME            | SIZE     | IDENTIFIER |
|----|-----------------------|-----------------|----------|------------|
| 0: | GUID_partition_scheme |                 | *1.0 TB  | disk0      |
| 1: | Apple_APFS_ISC        | Container disk1 | 524.3 MB | disk0s1    |
| 2: | Apple_APFS            | Container disk3 | 994.7 GB | disk0s2    |
| 3: | Apple_APFS_Recovery   | Container disk2 | 5.4 GB   | disk0s3    |


/dev/disk3 (synthesized):
#:

| #: | TYPE                  | NAME                    | SIZE      | IDENTIFIER |
|----|-----------------------|-------------------------|-----------|------------|
| 0: | APFS Container Scheme | -                       | +994.7 GB | disk3      |
|    |                       | Physical Store disk0s2  |           |            |
| 1: | APFS Volume           | Macintosh HD            | 11.3 GB   | disk3s1    |
| 2: | APFS Snapshot         | com.apple.os.update-... | 11.3 GB   | disk3s1s1  |
| 3: | APFS Volume           | Preboot                 | 7.1 GB    | disk3s2    |
| 4: | APFS Volume           | Recovery                | 1.0 GB    | disk3s3    |
| 5: | APFS Volume           | Data                    | 385.1 GB  | disk3s5    |
| 6: | APFS Volume           | VM                      | 6.4 GB    | disk3s6    |


/dev/disk4 (disk image):
#:

| #: | TYPE                  | NAME   | SIZE      | IDENTIFIER |
|----|-----------------------|--------|-----------|------------|
| 0: | GUID_partition_scheme |        | +760.2 MB | disk4      |
| 1: | Apple_HFS             | Claude | 760.2 MB  | disk4s1    |


/dev/disk5 (disk image):
#:

| #: | TYPE                  | NAME | SIZE      | IDENTIFIER |
|----|-----------------------|------|-----------|------------|
| 0: | GUID_partition_scheme |      | +892.3 MB | disk5      |
| 1: | Apple_HFS             | Trae | 892.3 MB  | disk5s1    |


/dev/disk6 (external, physical):
#:

| #: | TYPE                   | NAME | SIZE    | IDENTIFIER |
|----|------------------------|------|---------|------------|
| 0: | FDisk_partition_scheme |      | *4.0 GB | disk6      |
| 1: | Windows_NTFS           |      | 4.0 GB  | disk6s1    |


[sandrazilla@Mac ~ % sudo dc3dd if=/dev/disk6 of=dispositivoUSB.dd hash=sha256 log=dispositivoUSB.log
Password:

dc3dd 7.3.1 started at 2025-10-02 17:48:27 -0300
compiled options:
command line dc3dd if=/dev/disk6 of=dispositivoUSB.dd hash=sha256 log=dispositivoUSB.log
device size: 7831552 sectors (probed), 4,009,754,624 bytes
sector size: 512 bytes (probed)
391446528 bytes ( 373 M ) copied ( 10% ), 23 s, 16 M/s
```



{ [macOS Sequoia versión 15.6.1]

```
sandrazilla@Mac ~ % sudo dc3dd if=/dev/disk6 of=dispositivoUSB.dd hash=sha256 log=dispositivoUSB.log
Password:

dc3dd 7.3.1 started at 2025-10-02 17:48:27 -0300
compiled options:
command line dc3dd if=/dev/disk6 of=dispositivoUSB.dd hash=sha256 log=dispositivoUSB.log
device size: 7831552 sectors (probed), 4,009,754,624 bytes
sector size: 512 bytes (probed)
4009754624 bytes ( 3,7 G ) copied ( 100% ), 242 s, 16 M/s

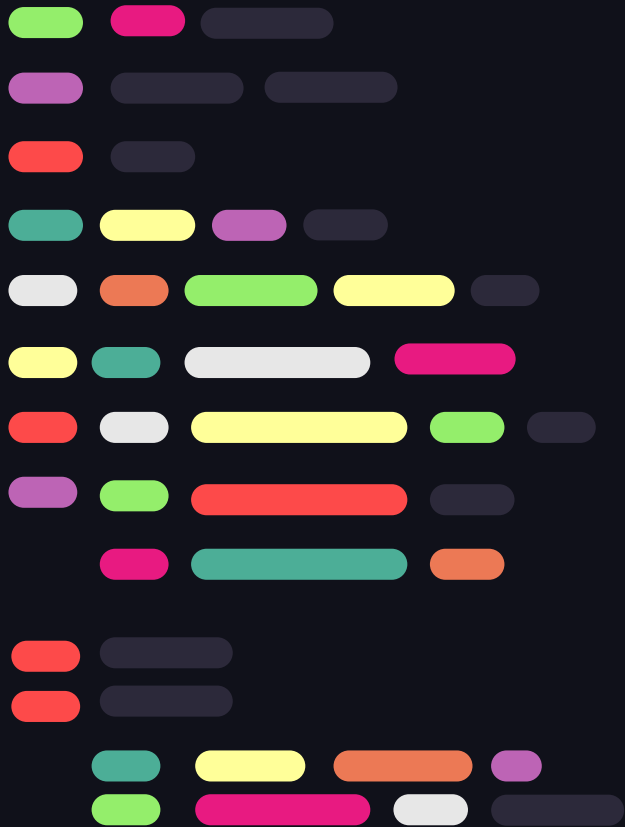
input results for device `/dev/disk6':
7831552 sectors in
0 bad sectors replaced by zeros
699f7f041bfb1a06884b98db526879496553d062fdbcb35a9bbce14d5bf923e8 (sha256)

output results for file `dispositivoUSB.dd':
7831552 sectors out

dc3dd completed at 2025-10-02 17:52:29 -0300
```

```
sandrazilla@Mac ~ % sha256sum dispositivoUSB.dd
699f7f041bfb1a06884b98db526879496553d062fdbcb35a9bbce14d5bf923e8 dispositivoUSB.dd
```





Guymager }

< Adquisición forense en Linux >



{ [Herramientas Forenses]

Guymager

GUI of Guymager 0.8.13 showing detected devices.

Terminal output (top right):

```
(nany@kali-fusion) ~/Desktop/dispositivoUSB  
$ sudo guymager  
[sudo] password for nany:  
Using default log file name /var/log/guymager.log  
StandardPaths: XDG_RUNTIME_DIR not set, defaulting to '/tmp/runtime-root'  
Using default cfg file name /etc/guymager/guymager.cfg
```

Serial	Linux device	Model	State	Size	Hidden areas	Bad sectors	Progress	Average speed [MB/s]	Time remaining
04160910001E4B9	/dev/sdb	General USB_Flash_Disk	Idle	4.0GB	unknown				
VMware NVME_0000	/dev/nvme0n1	VMware Virtual NVMe Disk	Idle	81.6GB	unknown				

Bottom panel (left):

- Size
- Sector size
- Image file
- Info file
- Current speed
- Started
- Hash calculation
- Source verification
- Image verification
- Overall speed (all acquisitions)

Bottom right: Computer icon.

{ [Herramientas Forenses Guymager] }

GUYMAGER 0.8.13

Devices Misc Help

Rescan

Serial nr.	Linux device	Model	State	Size	Hidden areas
041609100001E4B9	/dev/sdb	General USB_Flash_Disk	Idle	4.0GB	unknown
VMware NVME_0000	/dev/nvme0n1	VMware Virtual NVMe Disk	Idle	81.6GB	unknown

Size 4,009,754,624 bytes (3.73GiB / 4.01GB)
Sector size 512

Image file
Info file
Current speed
Started
Hash calculation
Source verification
Image verification
Overall speed (all acquisitions)

GUYMAGER 0.8.13

Devices Misc Help

Rescan

Serial nr.	Linux device	Model	State	Size	Hidden areas
041609100001E4B9	/dev/sdb	General USB_Flash_Disk	Idle	4.0GB	unknown
VMware NVME_0000	/dev/nvme0n1	VMware Virtual NVMe Disk	Idle	81.6GB	unknown

Acquire image
Clone device
Abort
Info



{ [Herramientas Forenses

Guymager]

Acquire image of /dev/sda

File format

☐ Linux dd raw image (file extension .dd or .xxx)

☒ Expert Witness Format, sub-format Guymager (file extension .Exx)

☒ Split image files

Split size 2047 MiB

Case number

Evidence number

Examiner

Description

Notes 7427EA2C39F2CD414EE194F7

Destination

Image directory ... /

Image filename (without extension)

Info filename (without extension)

Hash calculation / verification

☒ Calculate MD5 ☐ Calculate SHA-1 ☐ Calculate SHA-256

☐ Re-read source after acquisition for verification (takes twice as long)

☒ Verify image after acquisition (takes twice as long)

Cancel Duplicate image... Start

{ [Herramientas Forenses

Guymager]

Acquire image of /dev/sdb

File format

☒ Linux dd raw image (file extension .dd or .xxx) ☐ Split image files

☐ Expert Witness Format, sub-format Guymager (file extension .Exx) Split size 2047 MiB

Case number

Evidence number

Examiner

Description

Notes 041609100001E4B9

Destination

Image directory ... /home/nany/Desktop/dispositivoUSB/

Image filename (without extension) dispositivoUSB_2

Info filename (without extension) dispositivoUSB_2

Hash calculation / verification

☒ Calculate MD5 ☒ Calculate SHA-1 ☒ Calculate SHA-256

☐ Re-read source after acquisition for verification (takes twice as long)

☒ Verify image after acquisition (takes twice as long)

Cancel Duplicate image... Start

[Herramientas Forenses Guymager]

GUYMAGER 0.8.13

Devices Misc Help

Rescan

Serial nr.	Linux device	Model	State	Size	Hidden areas	Bad sectors	Progress	Average speed [MB/s]	Time remaining
041609100001E4B9	/dev/sdb	General USB_Flash_Disk	Running	4.0GB	unknown	0	11%	21.00	00:05:22
VMware NVME_0000	/dev/nvme0n1	VMware Virtual NVMe Disk	Idle	81.6GB	unknown				

Size	4,009,754,624 bytes (3.73GiB / 4.01GB)
Sector size	512
Image file	/home/nany/Desktop/dispositivoUSB/dispositivoUSB_2.dd
Info file	/home/nany/Desktop/dispositivoUSB/dispositivoUSB_2.info
Current speed	20.87 MB/s
Started	2. October 14:40:34 (00:00:42)
Hash calculation	MD5, SHA-1 and SHA-256
Source verification	off
Image verification	on
Overall speed (all acquisitions)	20.87 MB/s

[Herramientas Forenses Guymager]

GUYMAGER 0.8.13

Devices Misc Help

Rescan

Serial nr.	Linux device	Model	State	Size	Hidden areas	Bad sectors	Progress	Average speed [MB/s]	Time remaining
041609100001E489	/dev/sdb	General USB_Flash_Disk	Finished - Verified & ok	4.0GB	unknown	0	100%	33.84	
VMware NVME_0000	/dev/nvme0n1	VMware Virtual NVMe Disk	Idle	81.6GB	unknown				

Size 4,009,754,624 bytes (3.73GiB / 4.01GB)
Sector size 512
Image file /home/nany/Desktop/dispositivoUSB/dispositivoUSB_2.dd
Info file /home/nany/Desktop/dispositivoUSB/dispositivoUSB_2.info
Current speed
Started 2. October 14:40:34 (00:03:46)
Hash calculation MD5, SHA-1 and SHA-256
Source verification off
Image verification on
Overall speed (all acquisitions)

{ [Herramientas Forenses Guymager]

```
(nany@kali-fusion)-[~/Desktop/dispositivoUSB]
$ ls -lha
total 7.5G
drwxrwxr-x 2 nany nany 4.0K Oct  2 14:40 .
drwxr-xr-x 4 nany nany 4.0K Oct  2 11:57 ..
-rw-r--r-- 1 root root 3.8G Oct  2 12:38 dispositivoUSB.img
-rw-r--r-- 1 root root 3.8G Oct  2 14:43 dispositivoUSB_2.dd
-rw-r--r-- 1 root root 5.6K Oct  2 14:44 dispositivoUSB_2.info
-rw-r--r-- 1 root root 522 Oct  2 12:38 log_dispositivoUSB.txt
```



{ [Herramientas Forenses]

Guymager

```
(nany@kali-fusion)-[~/Desktop/dispositivoUSB]
$ cat dispositivoUSB_2.info

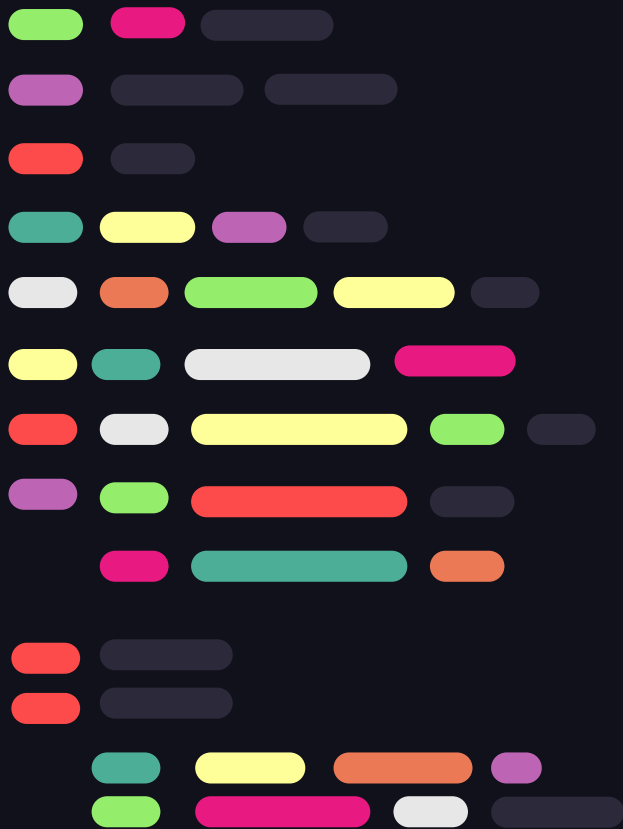
GUYMAGER ACQUISITION INFO FILE
```

```
Acquisition
=====
Linux device      : /dev/sdb
Device size       : 4009754624 (4.0GB)
Format            : Linux dd raw image - file extension is .dd
Image path and file name: /home/nany/Desktop/dispositivoUSB/dispositivoUSB_2.dd
Info path and file name: /home/nany/Desktop/dispositivoUSB/dispositivoUSB_2.info
Hash calculation   : MD5, SHA-1 and SHA-256
Source verification : off
Image verification  : on

No bad sectors encountered during acquisition.
State: Finished successfully

MD5 hash           : 96d3f913370ab0632e809be8954a74e8
MD5 hash verified source : --
MD5 hash verified image  : 96d3f913370ab0632e809be8954a74e8
SHA1 hash          : a78584a6048a70013a9821e33820e750f5f63a49
SHA1 hash verified source : --
SHA1 hash verified image  : a78584a6048a70013a9821e33820e750f5f63a49
SHA256 hash        : aecdd444fbc1bed12056833b78ebdeb36ab32390b84bbb8f083d88532ccff720
SHA256 hash verified source : --
SHA256 hash verified image : aecdd444fbc1bed12056833b78ebdeb36ab32390b84bbb8f083d88532ccff720
Image verification OK. The image contains exactly the data that was written.

Acquisition started : 2025-10-02 14:40:34 (ISO format YYYY-MM-DD HH:MM:SS)
Verification started: 2025-10-02 14:43:44
Ended               : 2025-10-02 14:44:21 (0 hours, 3 minutes and 46 seconds)
Acquisition speed   : 20.13 MByte/s (0 hours, 3 minutes and 10 seconds)
Verification speed   : 106.22 MByte/s (0 hours, 0 minutes and 36 seconds)
```



FTK



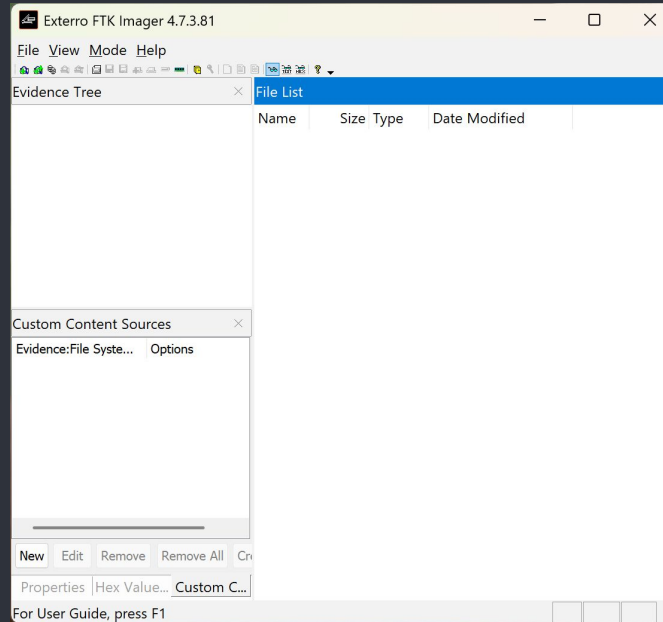
< Adquisición forense en Windows >





FTK Imager
Aplicación

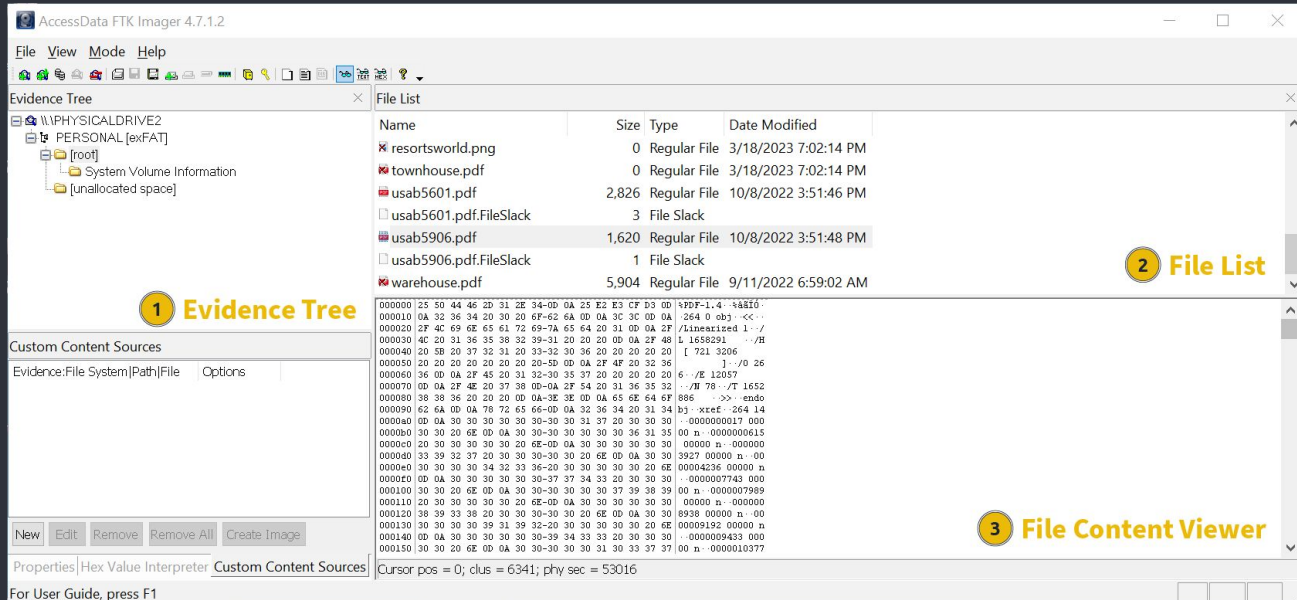
{ [Herramientas Forenses FTK] }



{ [Herramientas Forenses



FTK Imager
Aplicación





FTK Imager
Aplicación

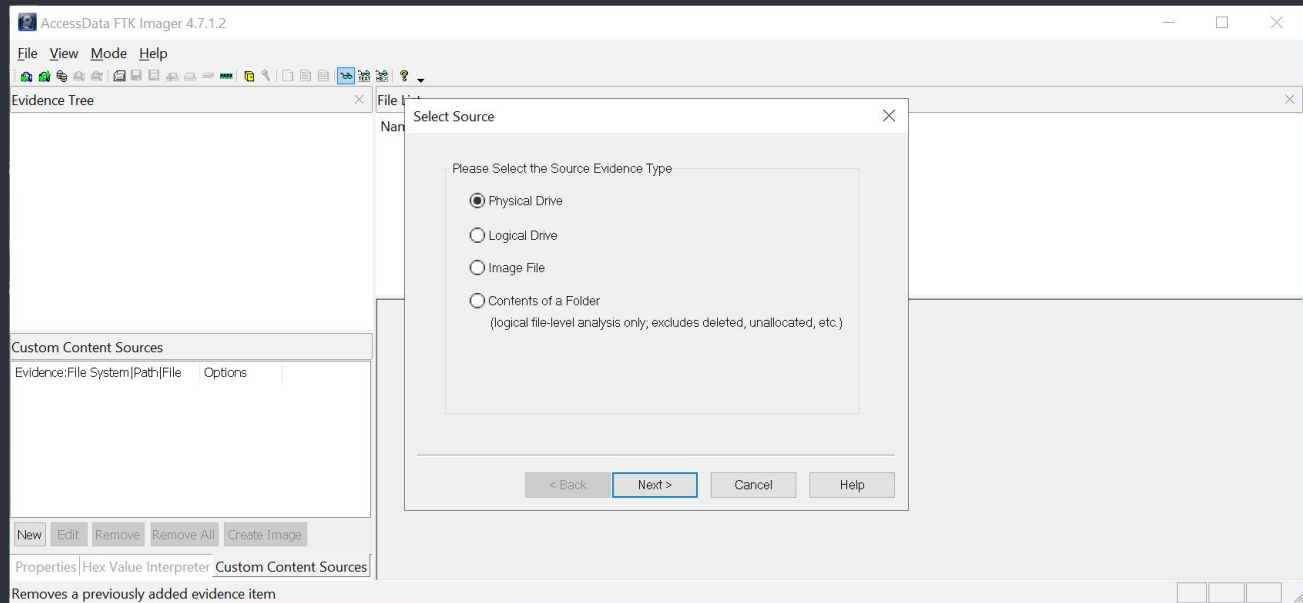
{ [Herramientas Forenses





FTK Imager
Aplicación

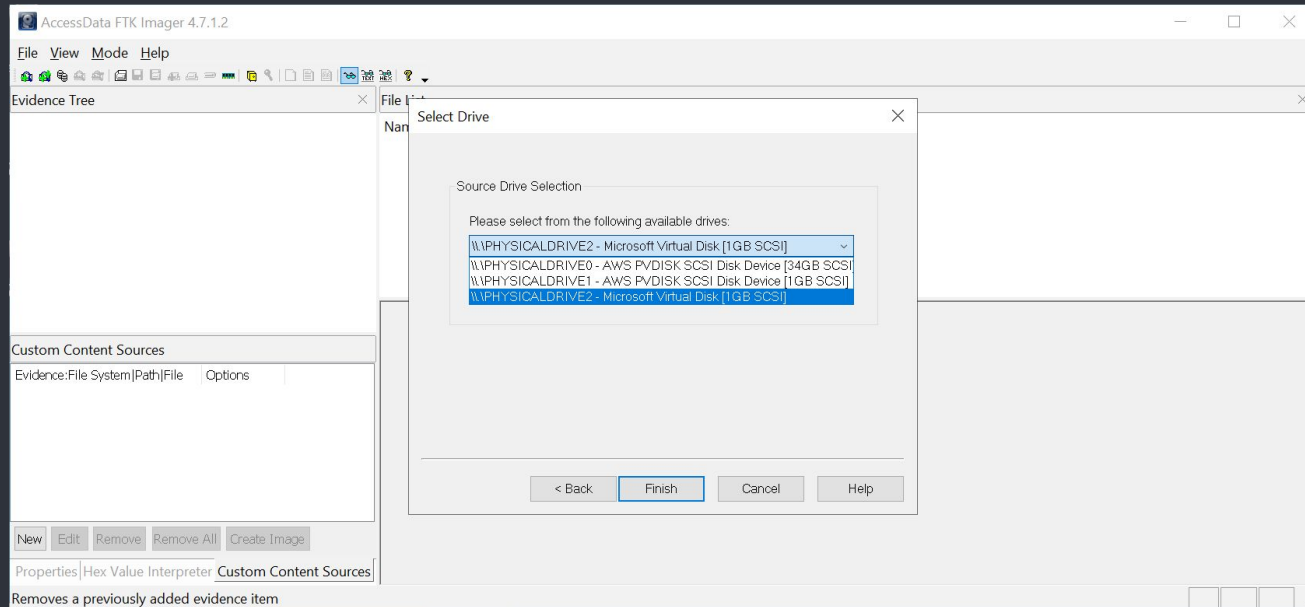
{ [Herramientas Forenses





FTK Imager
Aplicación

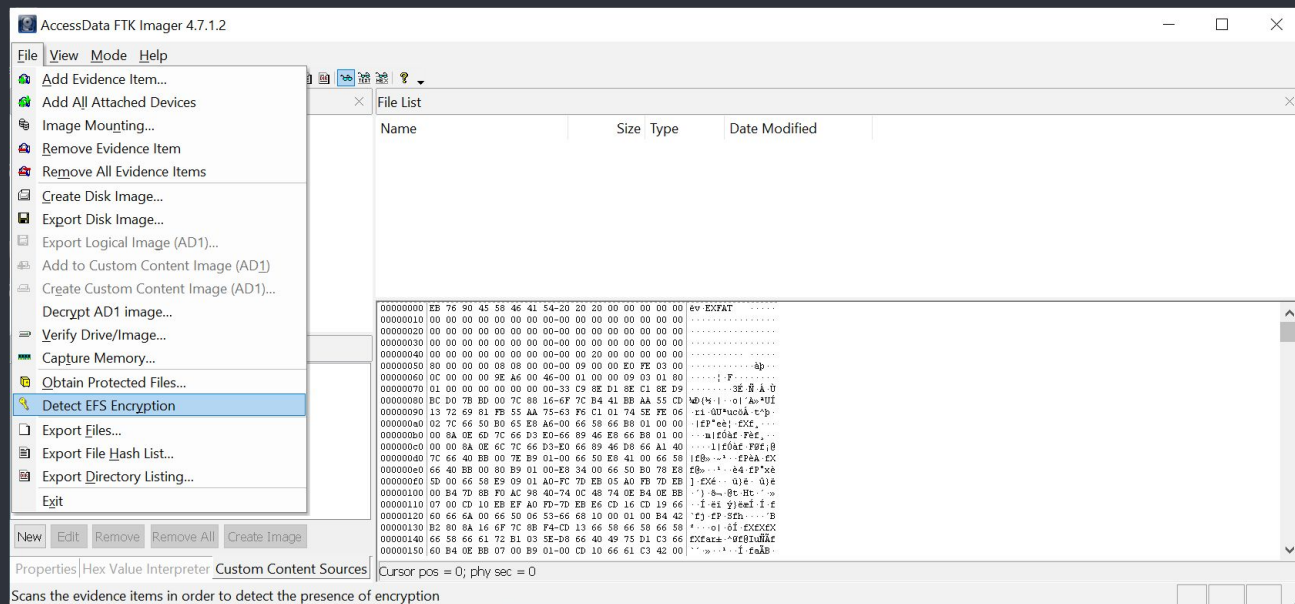
{ [Herramientas Forenses



{ [Herramientas Forenses



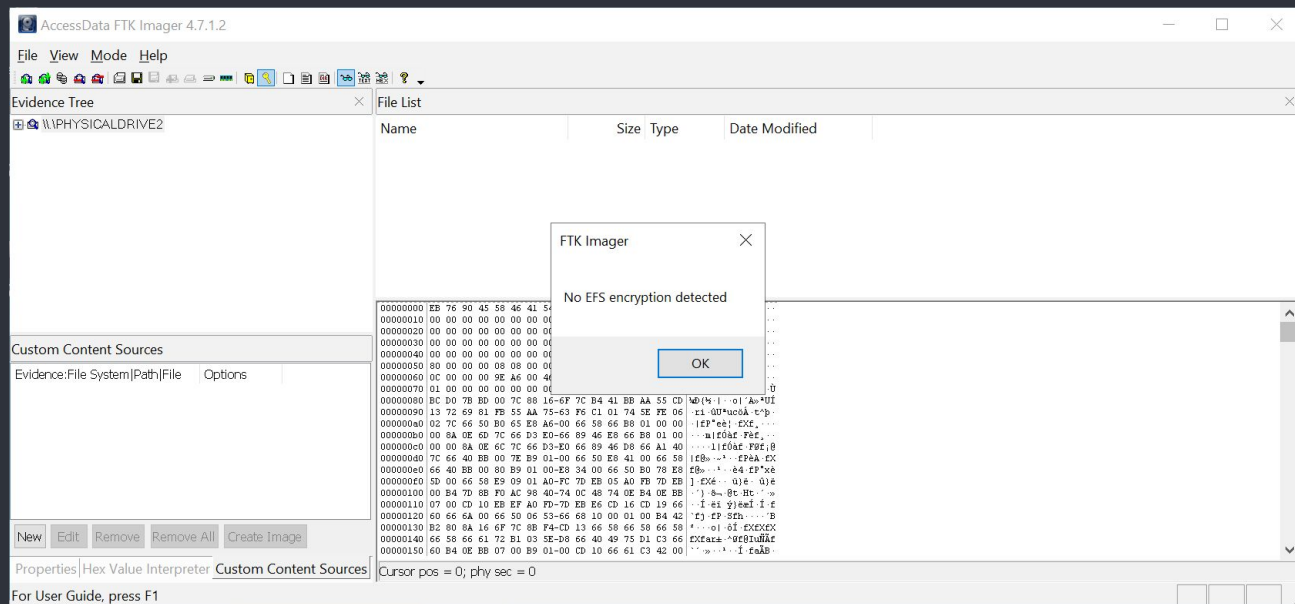
FTK Imager
Aplicación



{ [Herramientas Forenses

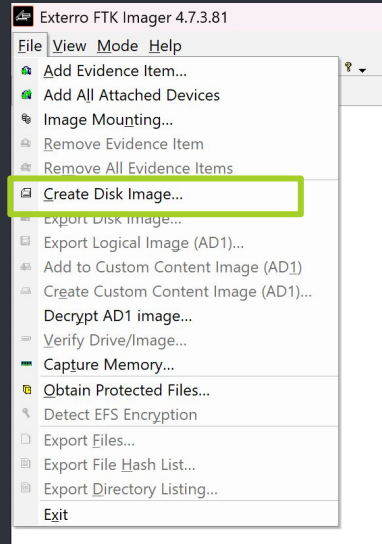


FTK Imager
Aplicación





{ [Herramientas Forenses

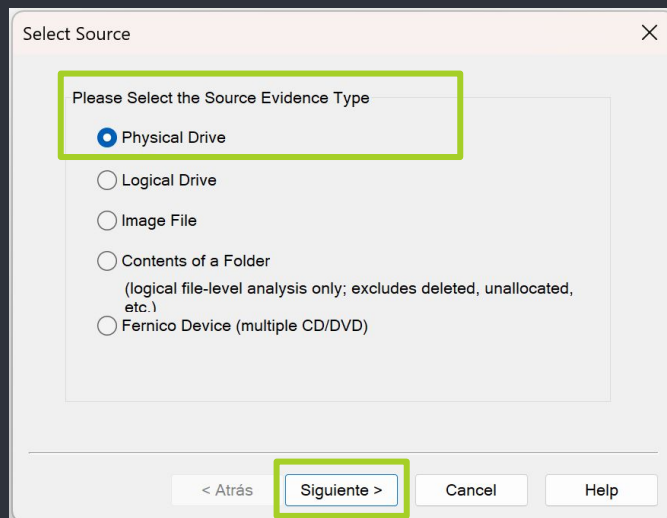


PASO 2: Creación de una imagen de disco forense con FTK Imager



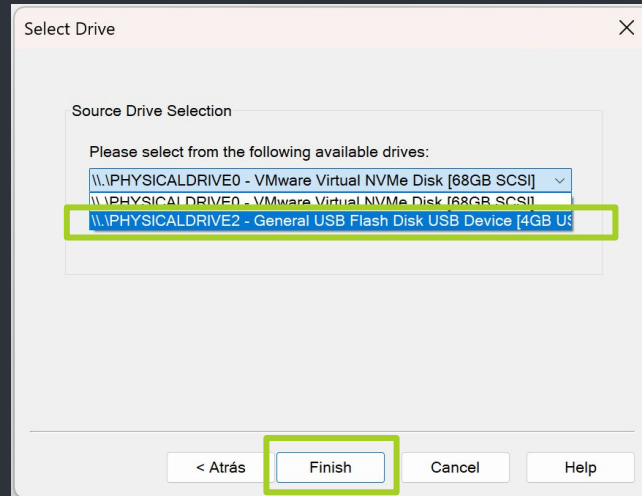


{ [Herramientas Forenses



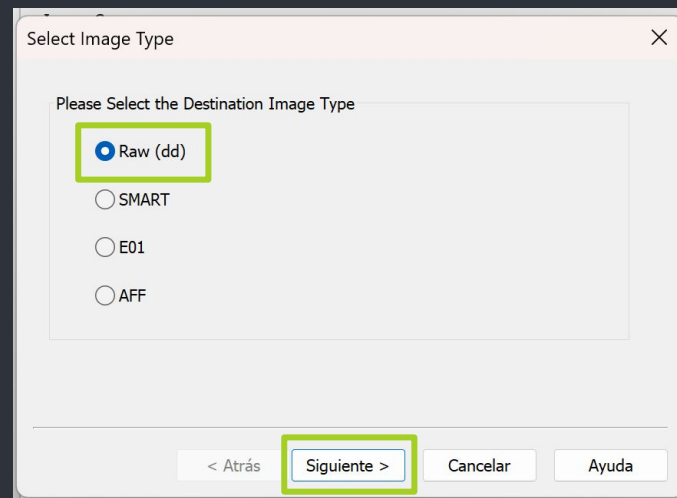
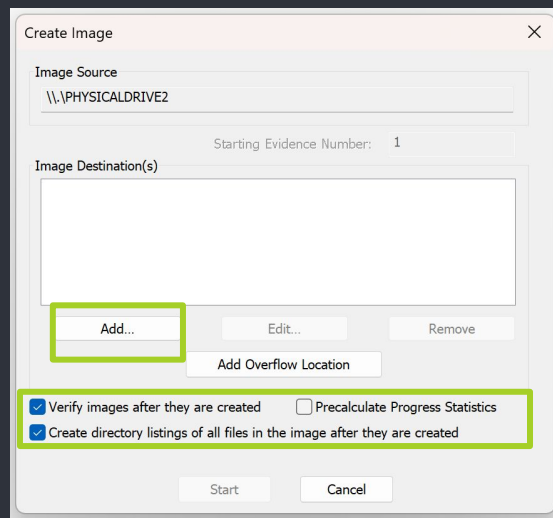


{ [Herramientas Forenses





{ [Herramientas Forenses





{ [Herramientas Forenses

Evidence Item Information

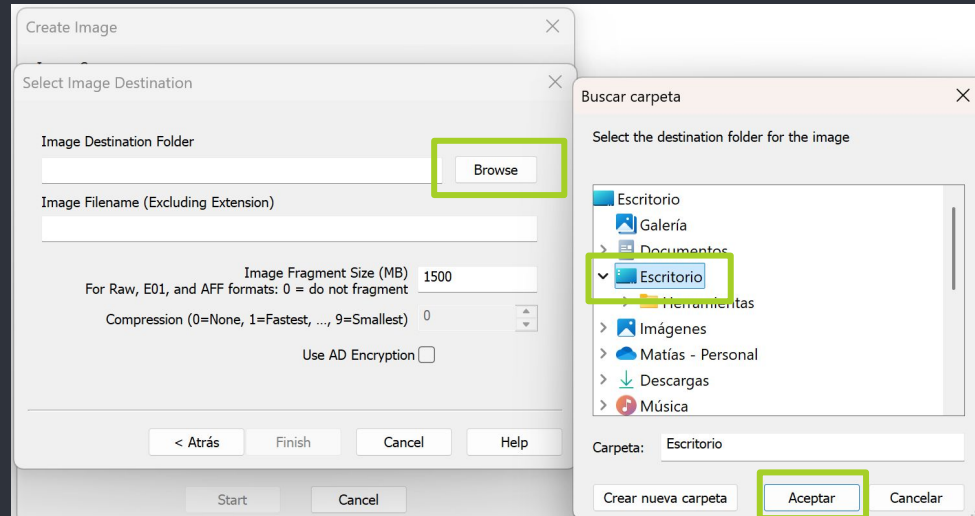
Case Number:	IFD-003456-2025
Evidence Number:	dispositivo1
Unique Description:	Pendrive
Examiner:	Sandra Zilla
Notes:	

< Atrás **Siguiente >** Cancel Help



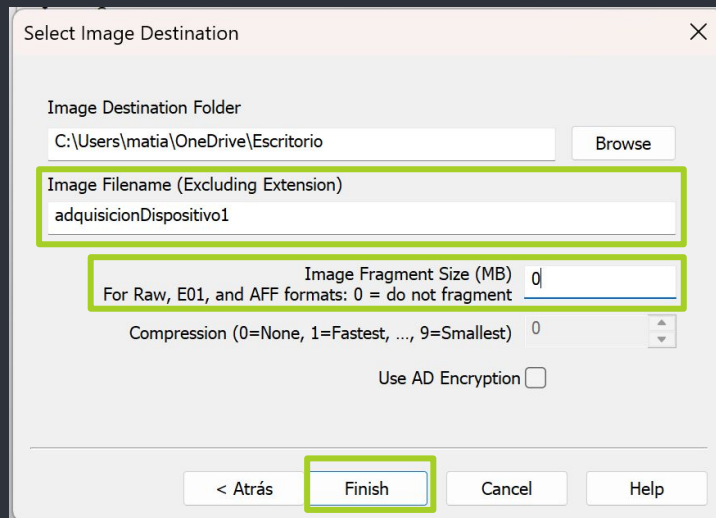


{ [Herramientas Forenses



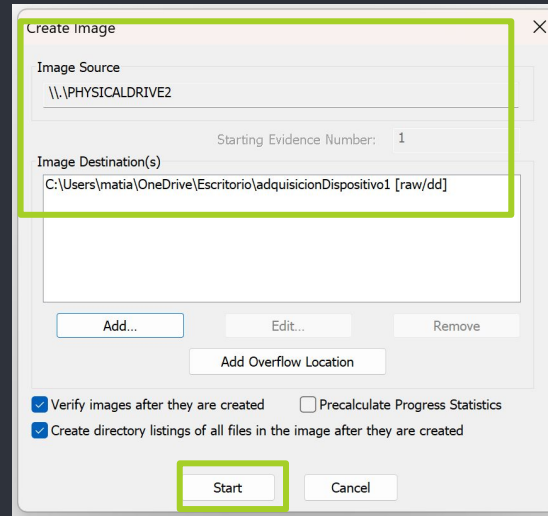


{ [Herramientas Forenses



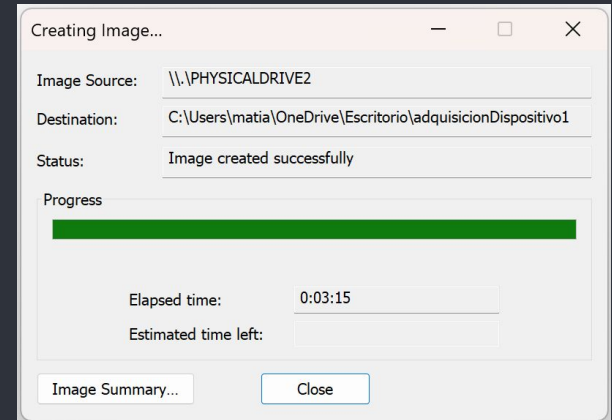
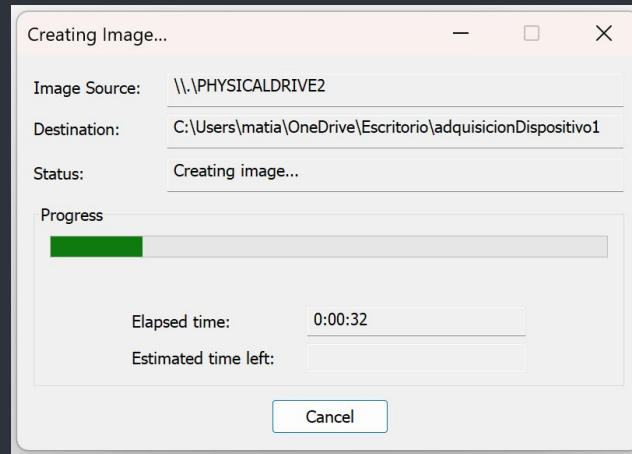


{ [Herramientas Forenses



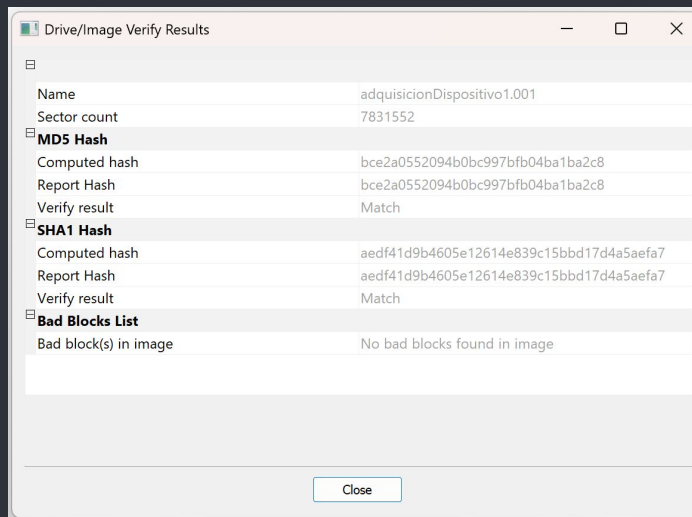


{ [Herramientas Forenses



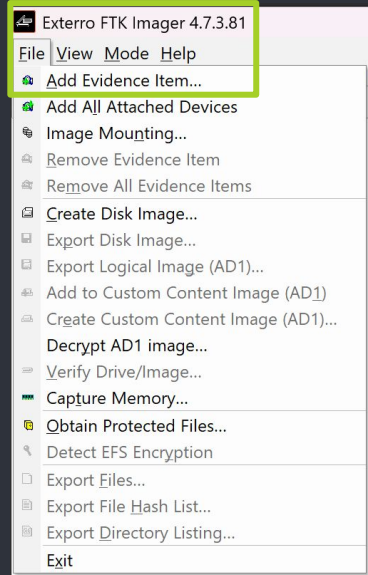


{ [Herramientas Forenses





{ [Herramientas Forenses

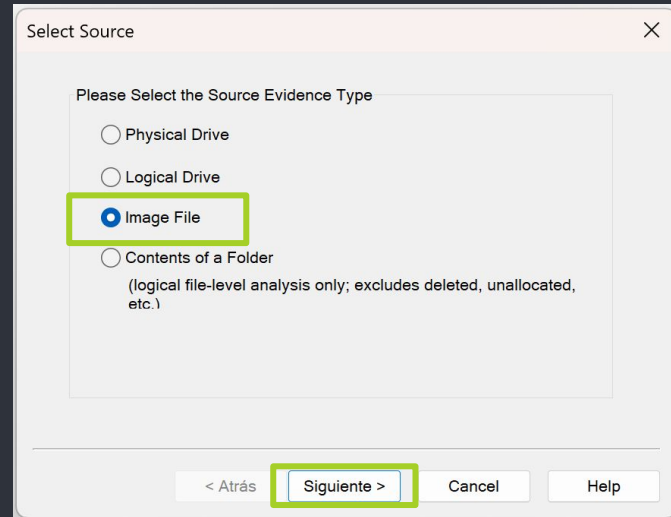


PASO 3: Agregar una imagen de disco forense y extracción de artefactos



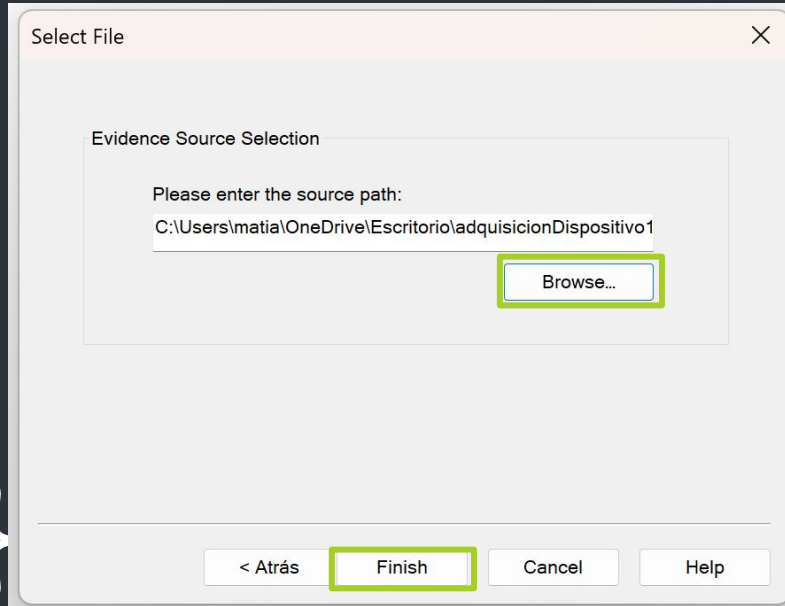


{ [Herramientas Forenses





[Herramientas Forenses



Fecha de modificación	Tipo	Tamaño
Herramientas		8/8/2025 16:08
adquisicionDispositivo1.001	Archivo 001	3.915.776 KB
adquisicionDispositivo1.001.csv	Archivo CSV	882 KB



{ [Herramientas Forenses

Exterro FTK Imager 4.7.3.81

File View Mode Help

Evidence Tree

- adquisicionDispositivo1.001
 - Partition 1 [3823MB]
 - NONAME [NTFS]
 - [orphan]
 - [root]
 - \$BadClus
 - \$Extend
 - \$Secure
 - \$UpCase
 - 345.docx
 - 345.pdf
 - 345.pdf
 - 345.pdf.backup
 - Arsenal-Image-Mounter-v3.11.307
 - bulk_extractor-1.5.0-windowsinstaller.exe
 - CellebriteReader.exe
 - hashmyfiles-x64
 - Ratool
 - SysinternalsSuite.zip
 - System Volume Information
 - [unallocated space]
 - Unpartitioned Space [basic disk]
 - [unallocated space]

Evidence Tree

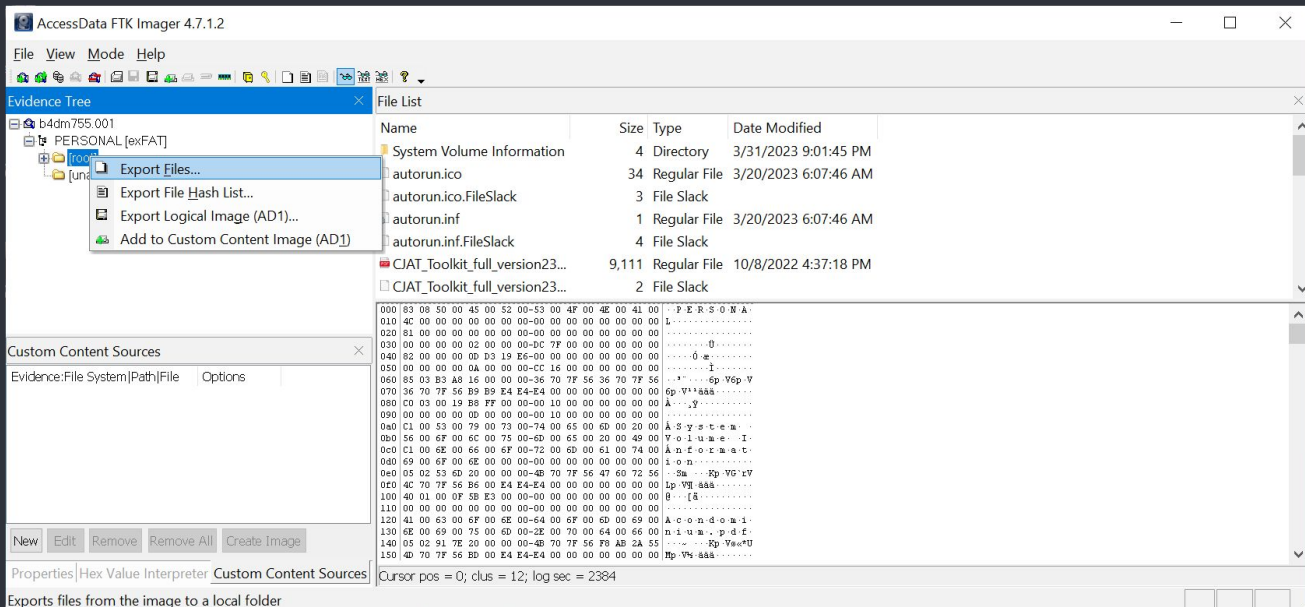
File List

Name	Size	Type	Date Modified
\$Extend	552 (1 KB)	Directory	1/1/2023 23:50:56
Arsenal-Image-Mounter-v3.11.307	304 (1 KB)	Directory	25/9/2025 20:50:01
hashmyfiles-x64	56 (1 KB)	Directory	25/9/2025 20:51:11
Ratool	256 (1 KB)	Directory	25/9/2025 20:49:58
System Volume Information	496 (1 KB)	Directory	2/1/2023 12:13:55
\$AttrDef	2.560 (3 KB)	Regular File	1/1/2023 23:50:56
\$BadClus	-	Regular File	1/1/2023 23:50:56
\$Bitmap	122.336 (120 KB)	Regular File	1/1/2023 23:50:56
\$Boot	8.192 (8 KB)	Regular File	1/1/2023 23:50:56
\$I30	4.096 (4 KB)	NTFS Index Allocation	30/9/2025 12:40:02
\$LogFile	20.041.728 (19.572 KB)	Regular File	1/1/2023 23:50:56
\$MFT	1.048.576 (1.024 KB)	Regular File	1/1/2023 23:50:56
\$MFTMirr	4.096 (4 KB)	Regular File	1/1/2023 23:50:56
\$Secure	56 (1 KB)	Regular File	1/1/2023 23:50:56
\$TXF_DATA	56 (1 KB)	NTFS Logged Utility Stream	30/9/2025 12:40:02
\$UpCase	131.072 (128 KB)	Regular File	1/1/2023 23:50:56
\$Volume	0 (0 KB)	Regular File	1/1/2023 23:50:56
345.docx	27.614 (27 KB)	Regular File	30/9/2025 11:51:37
345.pdf	27.614 (27 KB)	Regular File	30/9/2025 12:40:03
345.pdf	27.614 (27 KB)	Regular File	30/9/2025 12:01:12
345.pdf.backup	27.614 (27 KB)	Regular File	30/9/2025 11:51:37
bulk_extractor-1.5.0-windowsinstaller.exe	-	Regular File	30/9/2025 12:21:01
CellebriteReader.exe	1.038.597.472 (1.014.256 KB)	Regular File	15/6/2025 22:24:37
exploit.bin	0 (0 KB)	Regular File	30/9/2025 12:19:38
Nuevo Documento de texto.txt	6.787 (7 KB)	Regular File	30/9/2025 11:53:21
SysinternalsSuite.zip	174.194.607 (170.112 KB)	Regular File	30/9/2025 12:10:56

{ [Herramientas Forenses



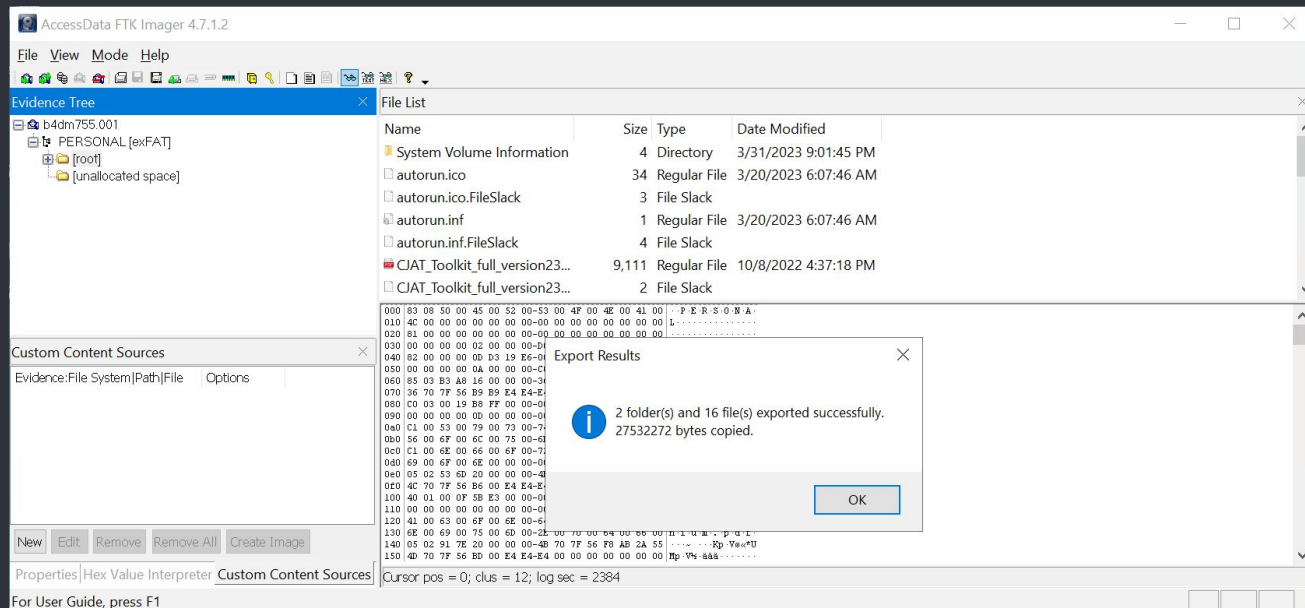
FTK Imager
Aplicación



{ [Herramientas Forenses



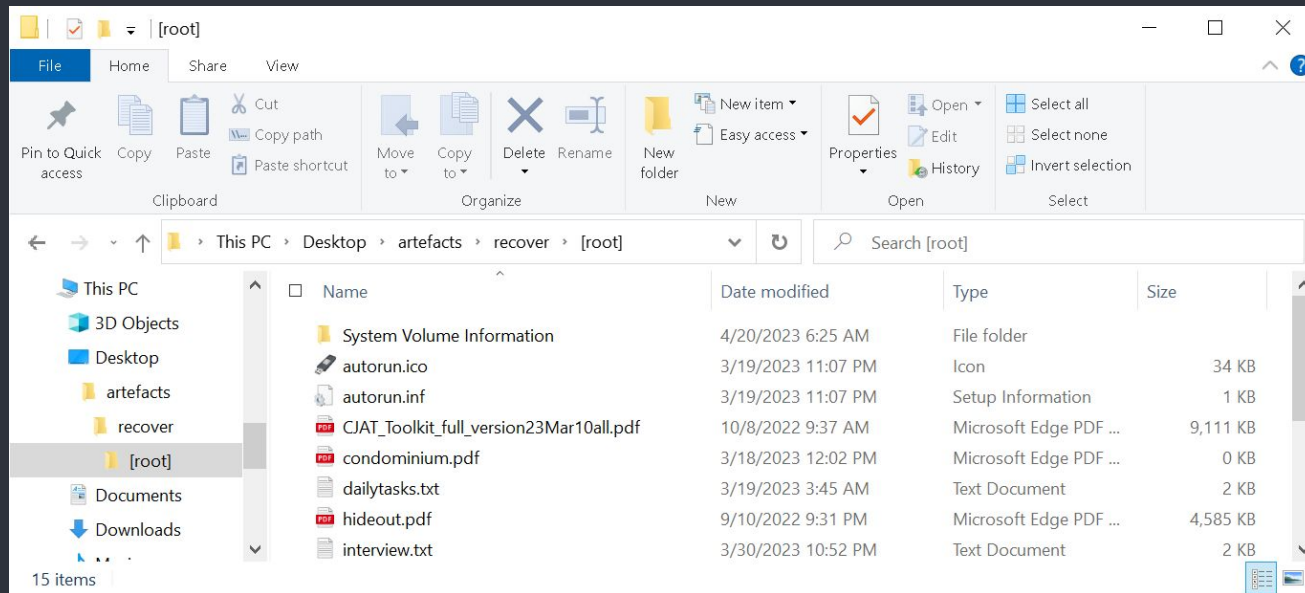
FTK Imager
Aplicación



{ [Herramientas Forenses

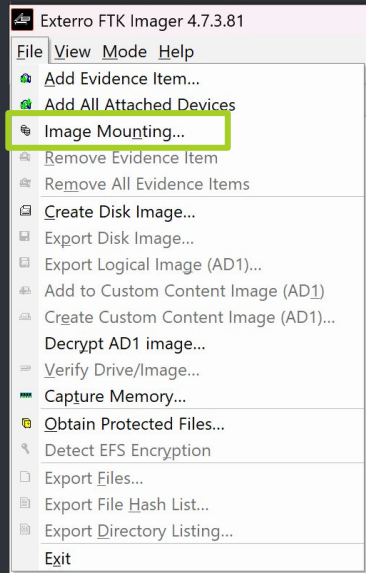


FTK Imager
Aplicación





{ [Herramientas Forenses

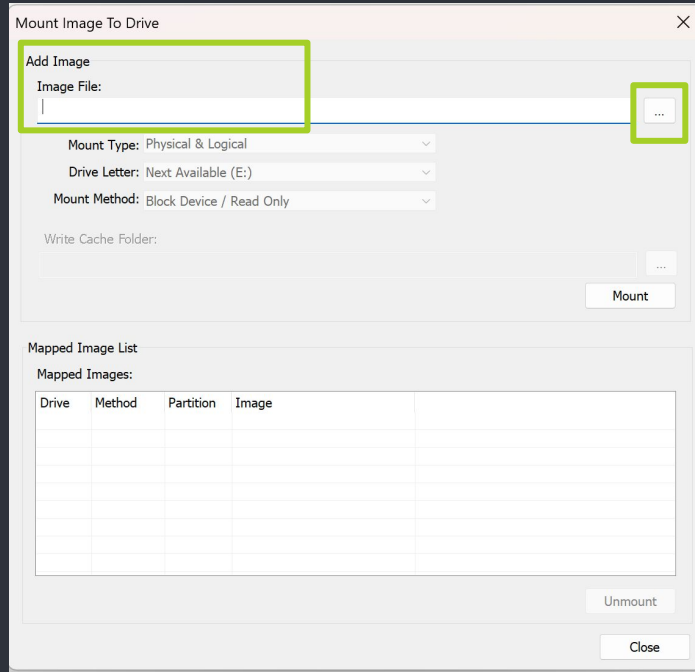


MONTAR una imagen de disco forense





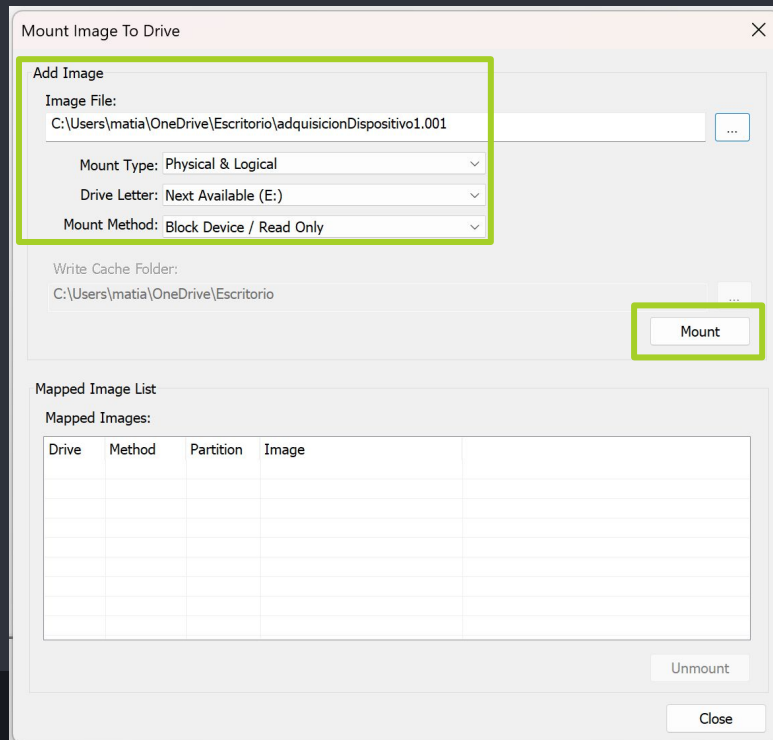
[Herramientas Forenses



Buscar en Escritorio				
	Fecha de modificación	Tipo	Tamaño	
	8/8/2025 16:08	Carpeta de archivos		
adquisicionDispositivo1.001	30/9/2025 10:15	Archivo 001	3.915.776 KB	
adquisicionDispositivo1.001.csv	30/9/2025 10:15	Archivo CSV	882 KB	



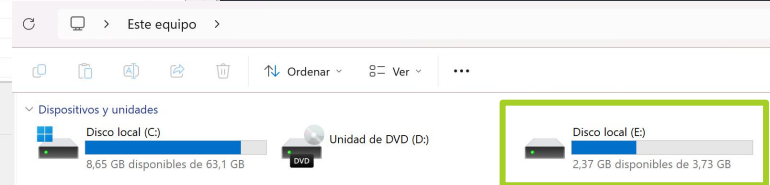
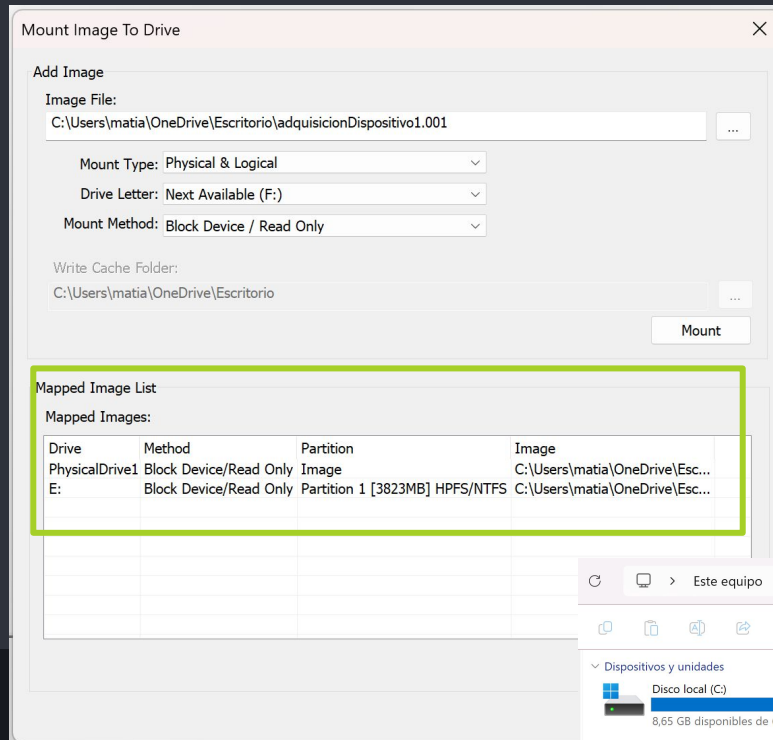
{ [Herramientas Forenses





FTK Imager
Aplicación

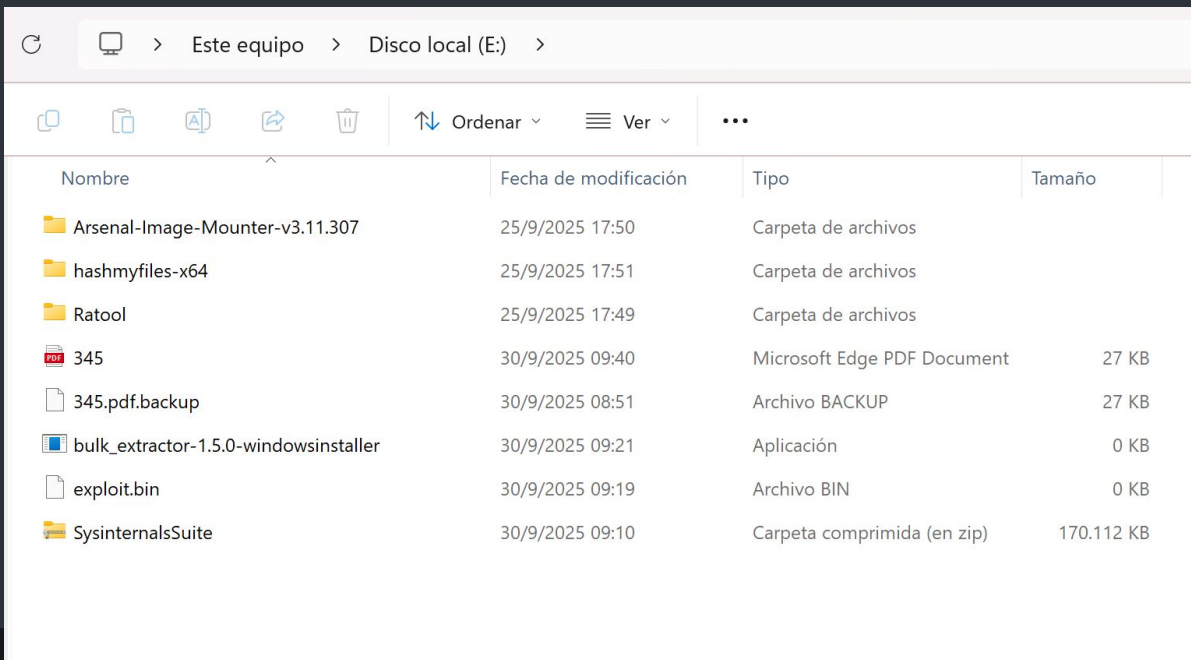
[Herramientas Forenses





FTK Imager
Aplicación

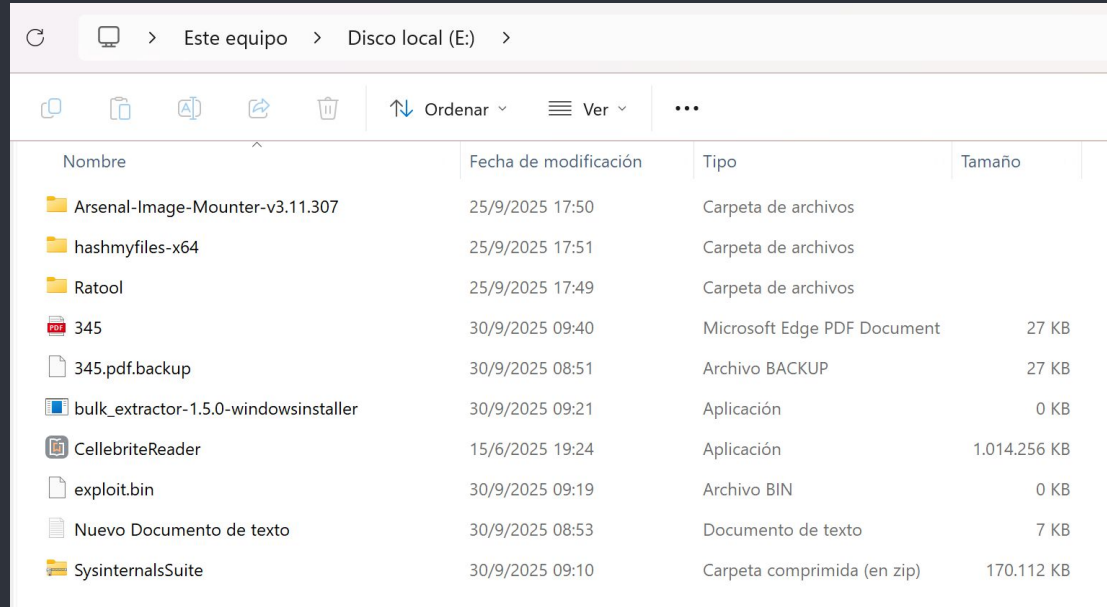
{ [Herramientas Forenses



Nombre	Fecha de modificación	Tipo	Tamaño
Arsenal-Image-Mounter-v3.11.307	25/9/2025 17:50	Carpeta de archivos	
hashmyfiles-x64	25/9/2025 17:51	Carpeta de archivos	
Ratool	25/9/2025 17:49	Carpeta de archivos	
345	30/9/2025 09:40	Microsoft Edge PDF Document	27 KB
345.pdf.backup	30/9/2025 08:51	Archivo BACKUP	27 KB
bulk_extractor-1.5.0-windowsinstaller	30/9/2025 09:21	Aplicación	0 KB
exploit.bin	30/9/2025 09:19	Archivo BIN	0 KB
SysinternalsSuite	30/9/2025 09:10	Carpeta comprimida (en zip)	170.112 KB



{ [Herramientas Forenses



Nombre	Fecha de modificación	Tipo	Tamaño
Arsenal-Image-Mounter-v3.11.307	25/9/2025 17:50	Carpeta de archivos	
hashmyfiles-x64	25/9/2025 17:51	Carpeta de archivos	
Ratool	25/9/2025 17:49	Carpeta de archivos	
345	30/9/2025 09:40	Microsoft Edge PDF Document	27 KB
345.pdf.backup	30/9/2025 08:51	Archivo BACKUP	27 KB
bulk_extractor-1.5.0-windowsinstaller	30/9/2025 09:21	Aplicación	0 KB
CellebriteReader	15/6/2025 19:24	Aplicación	1.014.256 KB
exploit.bin	30/9/2025 09:19	Archivo BIN	0 KB
Nuevo Documento de texto	30/9/2025 08:53	Documento de texto	7 KB
SysinternalsSuite	30/9/2025 09:10	Carpeta comprimida (en zip)	170.112 KB

```
1 { [Herramientas Forenses
2
3
4   para análisis de imágenes de disco]
5
6
7   < The Sleuth Kit (TSK) >
8   < Autopsy >
9   < EnCase >
10
11
12 }
13
14
```





Gracias

< Consultas >

szilla@info.unlp.edu.ar

einar@info.unlp.edu.ar



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